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COURSE NAME: Energy Analytics

PROJECT TITLE: Short-Term Hourly Electricity Load Forecasting for the CE Load Area

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INTRODUCTION

Problem Statement

Electricity demand must be continuously balanced with supply to ensure reliable power system operation. For large electric utility service territories such as the Commonwealth Edison (CE) load area, electricity demand exhibits strong hourly and seasonal variability, making short-term forecasting challenging. This project addresses the problem of accurately forecasting short-term hourly electricity load using historical data and demonstrates the usefulness of statistical predictive models in supporting efficient and reliable utility operations.

Motivation

Accurate short-term electricity load forecasting is essential for utilities and grid operators to make informed operational decisions. Poor forecasts can lead to inefficient generation scheduling, increased operational costs, and potential reliability risks during peak demand periods. With the increasing penetration of renewable energy and growing electricity demand, reliable forecasting methods are more important than ever. This motivates the need to evaluate different statistical forecasting techniques and identify models that are both accurate and practically useful for high-frequency load data.

Literature Review and Previous Work

Electricity load forecasting is a fundamental problem in power system planning and operation, as accurate short-term forecasts are essential for generation scheduling, grid reliability, and market operations. Traditionally, research and industry practices have focused on point forecasting methods. Hong and Fan (2016) provide a comprehensive tutorial review of probabilistic electric load forecasting, emphasizing the growing importance of uncertainty quantification due to renewable energy integration and increased system variability. Their work highlights that, despite advances in probabilistic approaches, accurate modeling of temporal patterns and seasonality remains the foundation of effective short-term load forecasting.

Taylor and McSharry (2008) evaluate short-term electricity load forecasting methods using European data and demonstrate that electricity demand exhibits strong daily and weekly seasonal patterns. Their study shows that statistical time-series models, particularly exponential smoothing methods designed to capture multiple seasonal cycles, can achieve high forecasting accuracy while remaining interpretable and computationally efficient. These findings motivate the use of classical time-series approaches such as ARIMA and ETS in this study to model the hourly PJM CE load data, which display similar periodic and seasonal characteristics.

Significance of the Problem

Short-term electricity load forecasting has a direct impact on power system planning, real-time grid operation, and energy market performance. Improved forecasting accuracy enables utilities to reduce reserve margins, optimize generation resources, and enhance overall grid reliability. From a broader

perspective, accurate demand forecasts support sustainable energy management by facilitating the efficient integration of renewable energy sources and reducing unnecessary energy waste. Consequently, the development of effective forecasting models for hourly electricity demand has significant economic, operational, and environmental importance.

Hypothesis

This study hypothesizes that hourly electricity load data exhibit strong daily, weekly, and annual seasonal patterns, which can be effectively captured using statistical time-series models. It is further hypothesized that, for short-term forecasting horizons, simple benchmark models such as Naïve or Exponential Smoothing methods may perform comparably to, or even outperform, more complex models such as ARIMA due to the high persistence and regularity present in electricity load data.

Proposed Statistical and Predictive Methods

To address the short-term electricity load forecasting problem, this study applies a set of classical statistical time-series forecasting methods. Benchmark models, including the Naïve and Seasonal Naïve approaches, are used to establish baseline forecasting performance. Exponential Smoothing (ETS) models are employed to capture level, trend, and seasonal components present in the hourly load data. In addition, Autoregressive Integrated Moving Average (ARIMA) models are applied to model temporal dependence and autocorrelation structures. These methods are selected due to their proven effectiveness, interpretability, and suitability for high-frequency electricity load forecasting. Model performance is evaluated using standard error metrics to identify the most accurate and practical forecasting approach.

Usefulness of the Proposed Predictive Models

The statistical forecasting models proposed in this study are well suited for short-term electricity load forecasting due to their ability to capture temporal dependence and strong seasonal patterns commonly observed in hourly demand data. Models such as Exponential Smoothing and ARIMA provide interpretable representations of level, trend, and seasonality, allowing system operators to understand the underlying drivers of electricity demand rather than relying on black-box predictions. This interpretability is particularly valuable for operational decision-making in power systems.

Furthermore, these models are computationally efficient and require only historical load data, making them practical for real-time implementation by utilities and grid operators. Accurate short-term forecasts generated by these methods can support generation scheduling, reserve planning, and congestion management, thereby improving system reliability and reducing operational costs. By comparing simple benchmark models with more structured statistical approaches, this study demonstrates how effective forecasting models can provide tangible operational benefits while maintaining simplicity and robustness.

PROJECT DESCRIPTION

To develop an effective electricity load forecasting model, the analysis was structured into a four-phase approach, as illustrated in Figure 1. This systematic framework ensures that the data are appropriately prepared, modeled, evaluated, and interpreted before drawing final conclusions.

Phase 1: Data Cleaning and Transformation
The first phase focused on preparing the raw hourly electricity load data for analysis. This included combining data from multiple years, filtering for the Commonwealth Edison (CE) load are**a**, and verifying overall data quality. Missing value checks were performed to ensure data completeness. In addition, a Box–Cox transformation was applied to stabilize variance and improve the suitability of the series for time-series modeling.

Phase 2: Data Modeling
In the second phase, multiple time-series forecasting models were implemented to capture the underlying patterns in electricity demand. Both simple benchmark models and more advanced statistical approaches were considered, including Naïve and Seasonal Naïve methods, drift-based forecasts, ARIMA models, and exponential smoothing techniques such as ETS. Seasonal-Trend decomposition using Loess (STL) was also applied to identify and analyze high-frequency seasonal patterns present in the hourly data.

Phase 3: Model Evaluation and Comparison
Model performance was evaluated using a train–test split strategy, with forecasts generated for a one-week horizon. Multiple accuracy metrics, including RMSE, MAE, MAPE, and MASE, were used to assess forecasting performance on the test set. Residual diagnostics and statistical tests were conducted to evaluate model adequacy and to identify any remaining autocorrelation or systematic patterns in the forecast errors.

Phase 4: Model Recommendation and Decision Making
In the final phase, the forecasting models were compared based on both quantitative accuracy metrics and qualitative diagnostic results. The most suitable model was selected by balancing forecast accuracy, model simplicity, and residual behavior. The selected model was then interpreted in the context of electricity demand behavior, and additional analyses were conducted to provide practical insights for short-term load forecasting and operational decision-making.

DATASET

The dataset used in this study consists of hourly electricity load measurements for the Commonwealth Edison (CE) load area, obtained from the PJM Interconnection data repository. PJM is a regional transmission organization that coordinates the movement of wholesale electricity across multiple states in the United States. The data cover a three-year period from 2022 to 2024 and represent the total metered electricity demand, measured in megawatts (MW), for the CE load zone.

Each observation includes a timestamp indicating the beginning of the hour and the corresponding electricity load value. The data are recorded at an hourly frequency, resulting in a high-resolution time

series suitable for detailed temporal analysis. Prior to analysis, the datasets from individual years were combined into a single continuous series and filtered to retain only the CE load area.

An initial data quality assessment confirmed that the dataset contains no missing values in either the timestamp or load variables, ensuring completeness for time-series modeling. The high frequency and large volume of observations make the dataset well suited for analyzing daily, weekly, and annual electricity consumption patterns, as well as for evaluating short-term electricity load forecasting models.

TASKS

Task 1

To understand the temporal behavior of hourly electricity load in the Commonwealth Edison (CE) load area, several exploratory visualizations were created. These plots help identify trends, seasonality, and unusual observations in the data.

Hourly Electricity Load (CE)

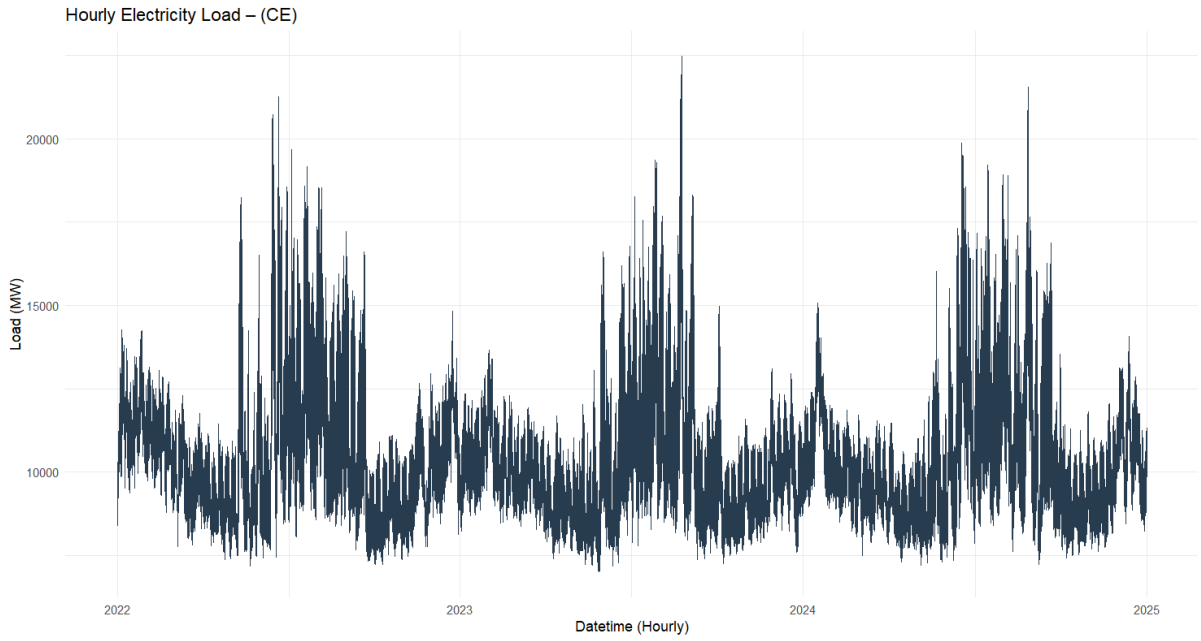


Figure1

Figure 1 displays the hourly electricity load for the (CE) region from 2022 through 2024 plotted against actual datetime values. The series exhibits substantial short-term variability due to the high frequency of hourly observations; however, several important patterns are clearly visible.

A strong annual seasonal pattern is evident, with electricity demand increasing significantly during summer and winter months and decreasing during spring and fall. The pronounced peaks observed in mid-year correspond to periods of high cooling demand, while elevated loads during winter months reflect heating requirements.

In addition to annual seasonality, the data exhibit repeated intraday and weekly fluctuations, visible as dense oscillations throughout the series. These reflect regular daily activity cycles and differences between weekday and weekend electricity usage.

Several sharp spikes in electricity demand are present, particularly during extreme summer and winter periods. These observations represent unusually high but realistic load values likely caused by extreme weather events such as heat waves or cold spells. These points are not treated as data errors but rather as important features of the demand process.

Overall, this figure indicates that the hourly electricity load series contains strong seasonal components at multiple time scales (daily, weekly, and annual), along with occasional extreme demand events. These characteristics motivate the use of decomposition techniques and seasonal forecasting models in subsequent analysis.

Daily Average Electricity Load – (CE), Year-wise

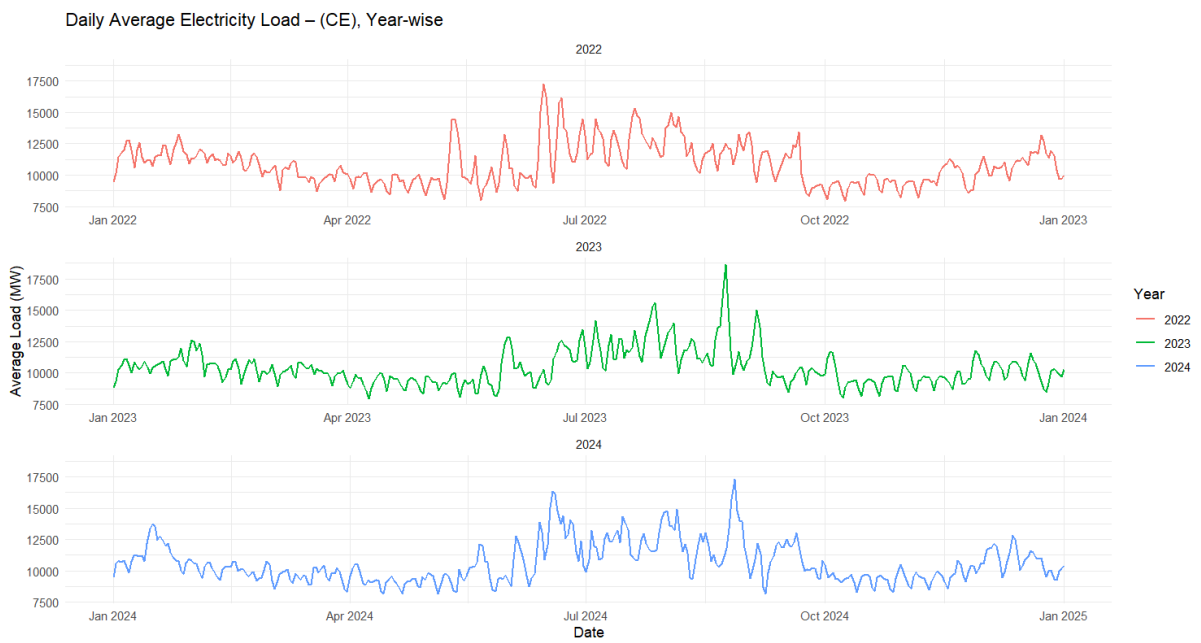


Figure 2

Figure 2 shows the daily average electricity load for the CE region, plotted separately for each year from 2022 to 2024. Aggregating the hourly data to daily averages reduces short-term volatility and allows clearer identification of long-term trends and seasonal behavior.

Across all three years, a consistent annual seasonal pattern is observed. Electricity demand increases during the summer months, reaching peak levels between June and August, which corresponds to elevated cooling demand. A secondary increase in load is visible during winter months, reflecting higher heating requirements. In contrast, spring and fall exhibit comparatively lower and more stable electricity demand.

While the overall seasonal pattern is consistent across years, differences in peak magnitudes are observed. For example, the summer peaks in 2023 and 2024 are slightly higher and more volatile than those in 2022, suggesting year-to-year variability likely driven by weather conditions. Several short-term spikes are also present within each year, indicating unusually high demand on specific days, potentially due to heat waves or cold spells.

Overall, this figure confirms the presence of strong annual seasonality and moderate interannual variability in electricity consumption, supporting the need for forecasting models that explicitly account for seasonal effects.

Average Hourly Load Pattern by Year

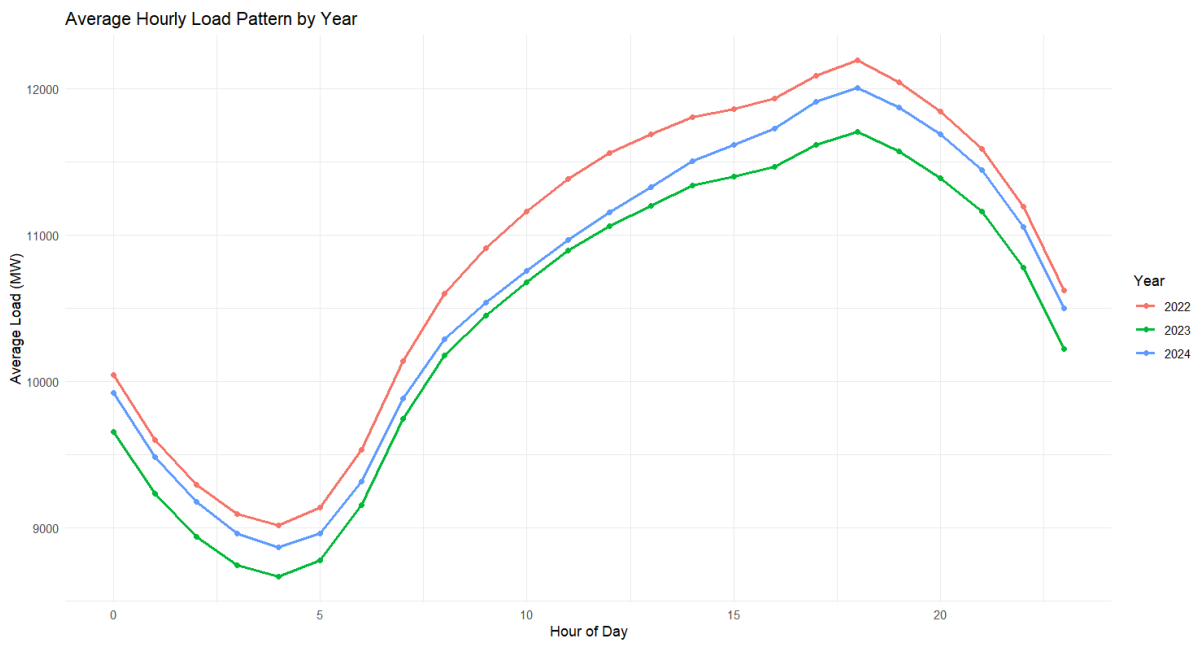


Figure 3

Figure 3 presents the average electricity load by hour of the day for the CE region, computed separately for each year from 2022 to 2024. This visualization highlights the intraday electricity consumption pattern while smoothing out day-to-day variability.

Across all three years, a clear and consistent daily cycle is observed. Electricity demand is highest during late evening and early night hours (around 18:00–20:00) and lowest during the early morning hours (approximately 03:00–05:00). This pattern reflects typical human activity, including residential energy use, commercial operations, and industrial demand throughout the day.

The similarity of hourly profiles across years indicates stable daily usage behavior, although slight differences in magnitude are present. For instance, 2022 exhibits marginally higher average load across most hours compared to 2023, while 2024 lies between the two. These differences likely reflect year-to-year variations in weather conditions and overall demand levels rather than structural changes in consumption behavior.

Overall, this figure confirms the presence of strong intraday seasonality in the electricity load data, supporting the use of forecasting models that explicitly account for daily and weekly seasonal effects.

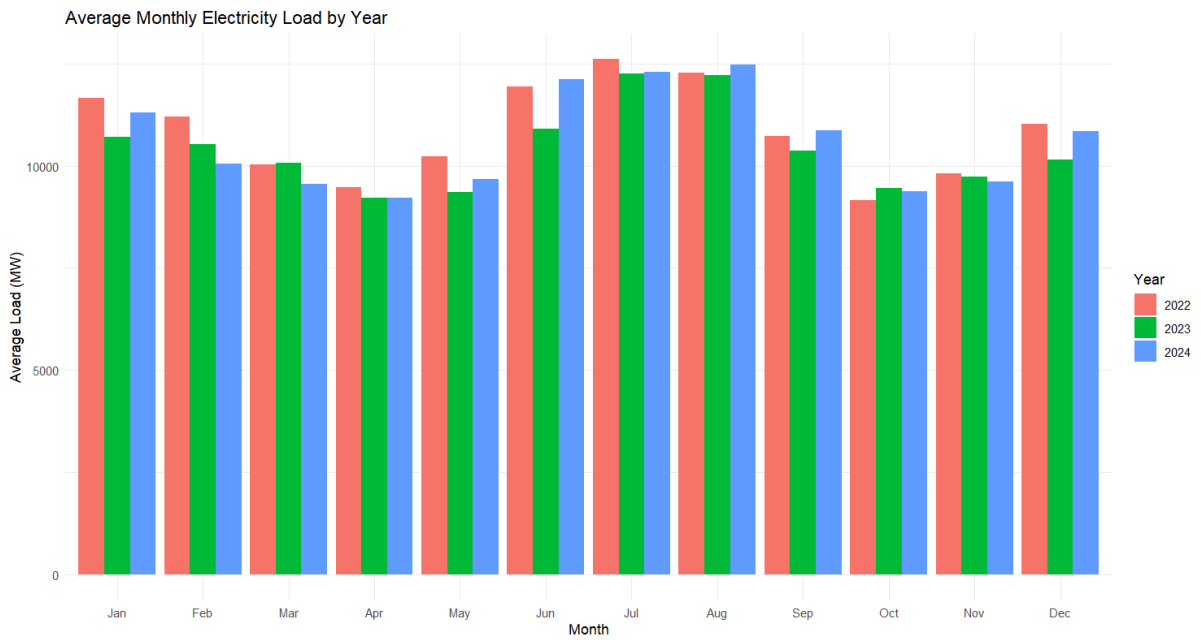


Figure 4

Figure 4 displays the average monthly electricity load for the CE region, compared across the years 2022, 2023, and 2024. Aggregating the data at the monthly level smooths short-term fluctuations and highlights long-term seasonal patterns in electricity demand.

A strong annual seasonal cycle is clearly visible across all three years. Electricity demand is highest during the summer months, particularly in July and August, reflecting increased cooling requirements. Elevated demand is also observed during winter months such as January and December, corresponding to heating needs. In contrast, spring and fall months (April, May, October) exhibit lower and more stable electricity consumption.

While the overall seasonal shape is consistent, moderate interannual variability is evident. For example, summer demand in 2022 appears slightly higher than in 2023, while 2024 shows comparable or marginally higher loads in late summer and winter months. These differences likely reflect variations in weather intensity and economic activity across years.

Overall, this figure confirms pronounced annual seasonality in electricity consumption and supports the need for forecasting models that account for seasonal effects at multiple time scales.

Task 2

The dataset was examined for missing values in both the timestamp (datetime) and electricity load (mw) variables. A systematic check using column-wise and total missing value counts revealed that no missing values are present in the dataset.

Specifically, the total number of missing observations in the dataset is zero. Since the dataset is complete, no imputation or deletion was required, and all observations were retained for subsequent analysis.

```
> # missing values
> colSums(is.na(CE_data))
datetime       mw
       0        0
> sum(is.na(CE_data))
[1] 0
```

Figure 5

Task 3

Stl decomposition

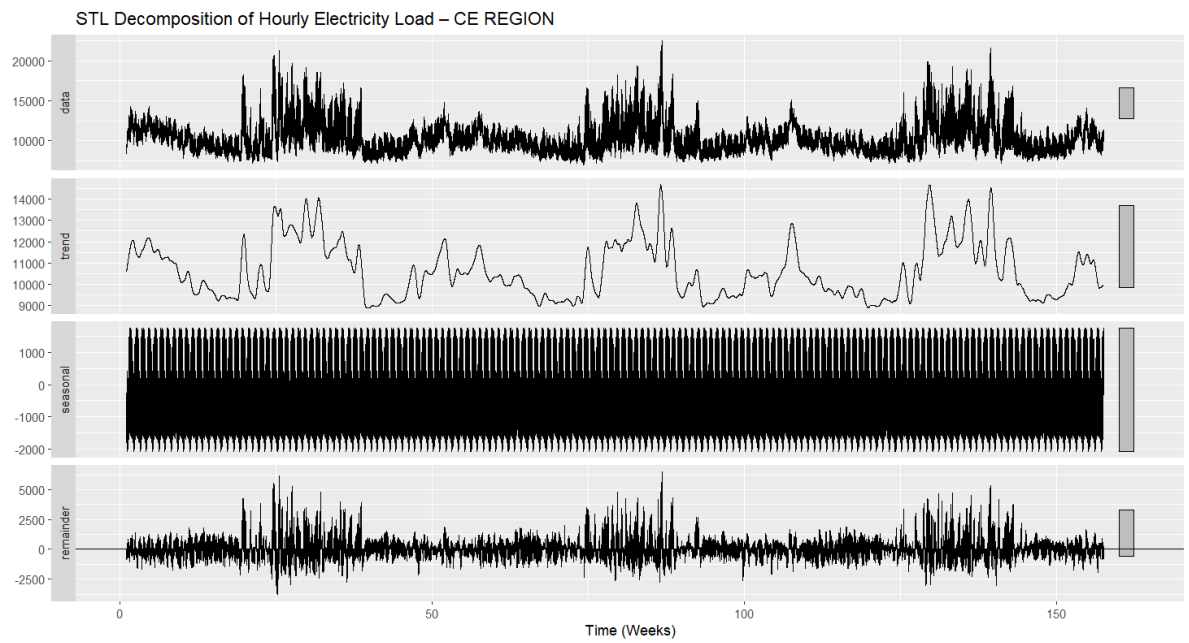


Figure 6

Figure 6 presents the STL (Seasonal and Trend decomposition using Loess) decomposition of the hourly electricity load for the CE region. The x-axis represents time measured in **weekly seasonal cycles**, since the series was modeled as an hourly time series with a weekly seasonal frequency of 168

hours. Approximately 150 weeks of data are shown, corresponding to nearly three years of observations.

The **data (observed) component** shows the original hourly electricity load series. High short-term variability is evident due to hourly fluctuations, along with repeated periods of elevated demand during summer and winter seasons. Several sharp peaks are visible, indicating extreme demand events.

The **trend component** captures the smooth long-term evolution of electricity demand. The trend increases during summer and winter periods, reflecting higher cooling and heating demand, and decreases during spring and fall. This confirms the presence of long-term seasonal trends observed in the exploratory analysis.

The **seasonal component** displays a highly regular and stable pattern over time, with consistent amplitude across the entire series. This indicates strong and persistent intraday and weekly seasonality in electricity consumption, driven by daily activity cycles and weekday–weekend effects. The stability of this component suggests that seasonal behavior does not change significantly across years.

The **remainder component** represents irregular fluctuations not explained by the trend or seasonal components. Most values are centered around zero, indicating that STL successfully captures the primary structure of the data. However, several large spikes are present in the remainder, particularly during periods of extreme summer and winter demand.

Graph: *Trend Component from STL – (CE)*

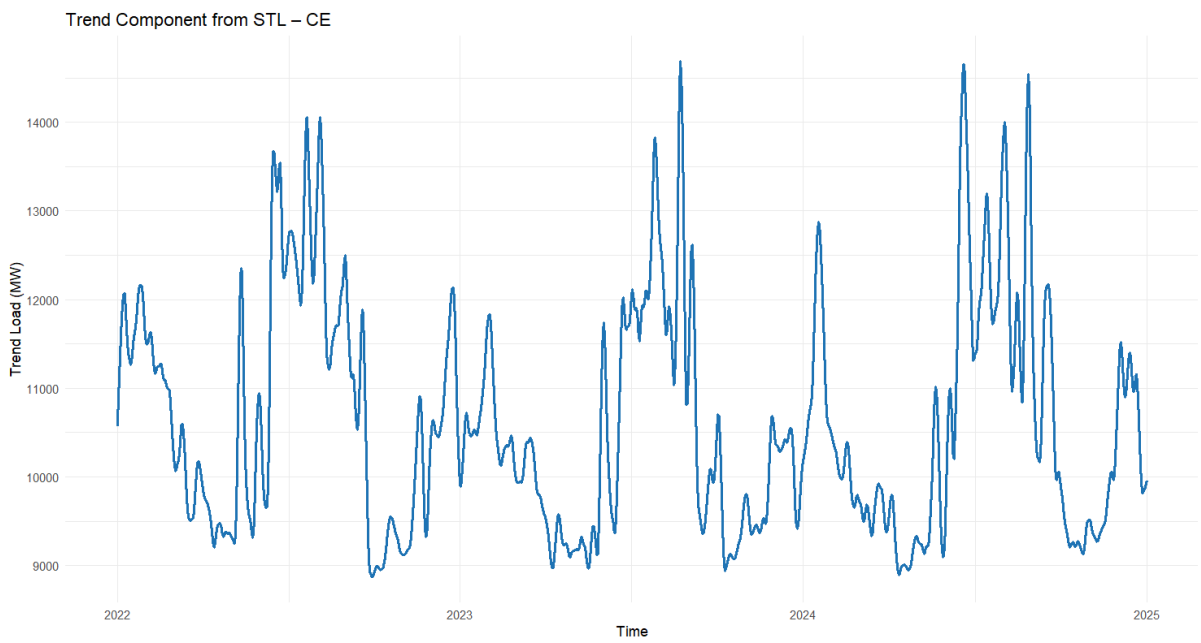


Figure 7

The trend component extracted from the STL decomposition shows smooth long-term variations in electricity demand over the three-year period. The trend exhibits recurring increases during summer and winter months, corresponding to higher cooling and heating demand, and lower levels during spring and fall. This behavior confirms the presence of a clear trend component, consistent with the patterns identified in the exploratory analysis in Task 1.

Seasonal Component (One-Week View)

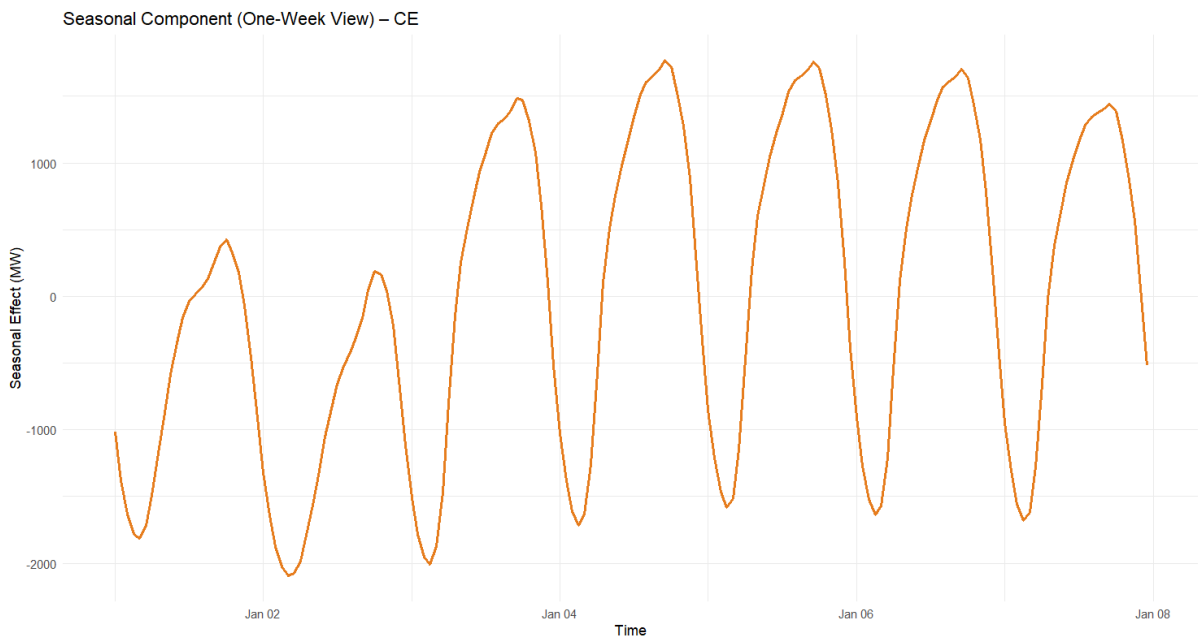


Figure 8

To improve interpretability of the seasonal component, a one-week window was visualized. The seasonal component reveals a strong and stable hourly cycle, reflecting daily and weekly usage patterns. The consistent shape and amplitude of this component indicate pronounced seasonality in electricity consumption, supporting the daily and weekly seasonal patterns observed in Task 1.

Remainder Component Plot

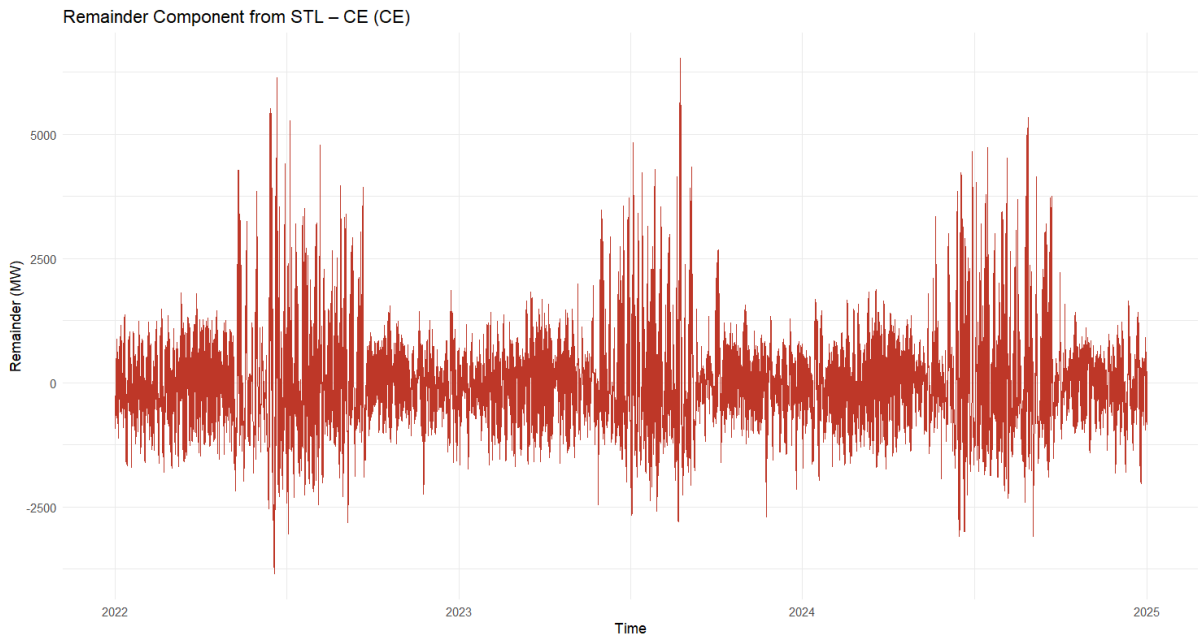


Figure 9

The remainder component represents the irregular variation in the time series after removing the trend and seasonal effects. Most of the remainder values fluctuate around zero, indicating that the majority of systematic patterns have been successfully captured by the decomposition. However, several isolated spikes are visible, suggesting unusual demand events that were not easily detectable in the raw time series. These spikes likely correspond to extreme weather conditions or sudden demand surges.

Task 4

```
> lambda <- BoxCox.lambda(CE_ts)
> lambda
[1] -0.9999242
>
```

Figure 10

Transformation of the Time Series

The hourly electricity load series was examined to determine whether a transformation was required prior to modeling. Visual inspection of the raw data revealed that the variability of the series increases during periods of high electricity demand, particularly in summer and winter months. This suggests non-constant variance, which may violate the assumptions of many time series models such as ARIMA and ETS.

To further assess the suitability of a transformation, a Box–Cox analysis was conducted. The estimated Box–Cox parameter was approximately −1, indicating that a strong variance-stabilizing transformation may be appropriate for the data. Accordingly, a Box–Cox transformation was applied to stabilize variance and improve the suitability of the series for forecasting models. Due to the

strength of the transformation, the transformed values appear compressed, which is expected and does not adversely affect model estimation or forecasting performance.

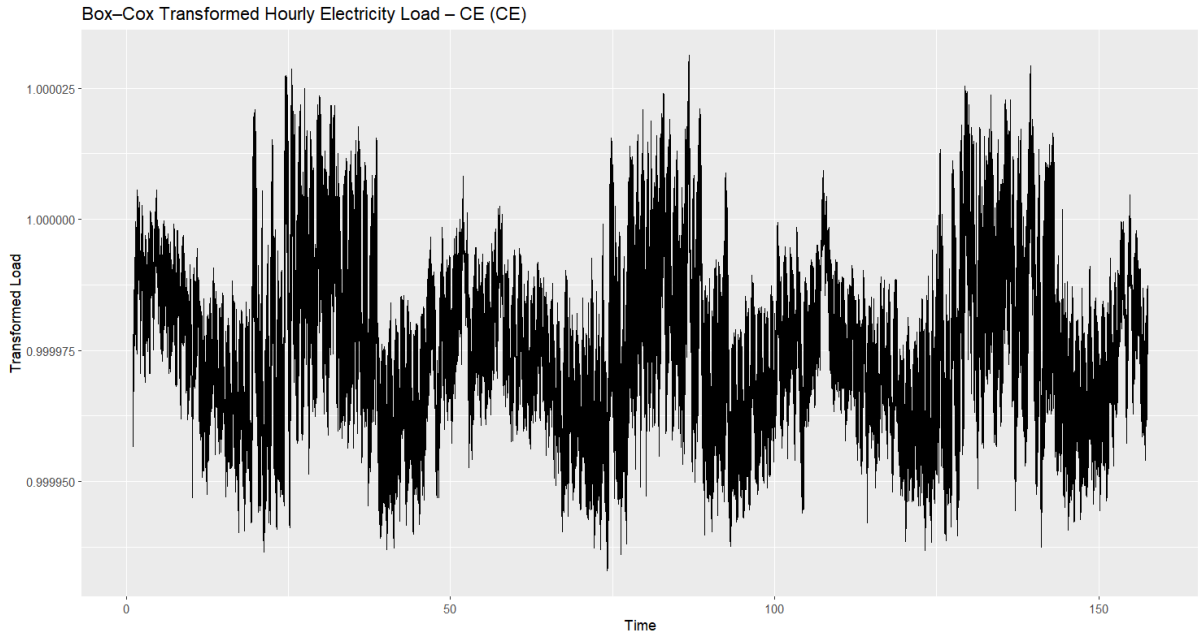


Figure 11 (Time in weeks)

A Box–Cox transformation was applied to the hourly electricity load series. The estimated Box–Cox parameter was approximately −1, indicating that a strong variance-stabilizing transformation may be appropriate for the data. The transformed series was used in subsequent modeling to improve forecast performance.

TASK 5

Different forecasting approaches were implemented to model and predict hourly electricity load for the CE (ComEd) zone.

Naive Method

The Naive forecasting method assumes that the forecast for all future periods is equal to the most recent observed value. Despite its simplicity, this method often performs well for short-term forecasting when the time series exhibits strong persistence.

Seasonal Naive Method

The Seasonal Naïve method extends the Naïve approach by accounting for seasonality. For hourly electricity load data with weekly seasonality, the forecast for a given hour is set equal to the observed

value from the same hour in the previous week. This model serves as an important baseline for seasonal time series.

STL + ETS Method

To address the high seasonal frequency of the hourly data, seasonal forecasting was performed using STL decomposition followed by an ETS model (STL+ETS). The STL procedure decomposes the time series into trend, seasonal, and remainder components, after which an ETS model is applied to the seasonally adjusted series. This approach allows seasonal patterns to be handled effectively even when the seasonal frequency exceeds the limitations of standard ETS models.

ARIMA Method

An ARIMA (AutoRegressive Integrated Moving Average) model was fitted using automatic model selection. The auto.arima() function was used to identify appropriate autoregressive, differencing, and moving average orders while accounting for seasonality. ARIMA models are widely used for time series forecasting due to their flexibility and strong theoretical foundation.

```
                     MASE         RMSE          MAE
Naive          0.8252737 8.721163e-06 7.564076e-06
Seasonal_Naive 1.3357133 1.355371e-05 1.224253e-05
STL_ETS        6.1443362 5.858075e-05 5.631614e-05
ARIMA          1.3357133 1.355371e-05 1.224253e-05
                     MAPE
Naive          0.0007564292
Seasonal_Naive 0.0012242891
STL_ETS        0.0056317668
ARIMA          0.0012242891
```

```
> best_model <- rownames(perf_table)[which.min(perf_table[, "RMSE"])]
> best_model
[1] "Naive"
```

Train–Test Split and Evaluation Procedure

To evaluate forecasting performance, a train–test split approach was used. The final one-week period (168 hours) of the dataset was reserved as the test set, while the remaining observations were used for model training. All models generated forecasts for the same horizon to ensure a fair comparison.

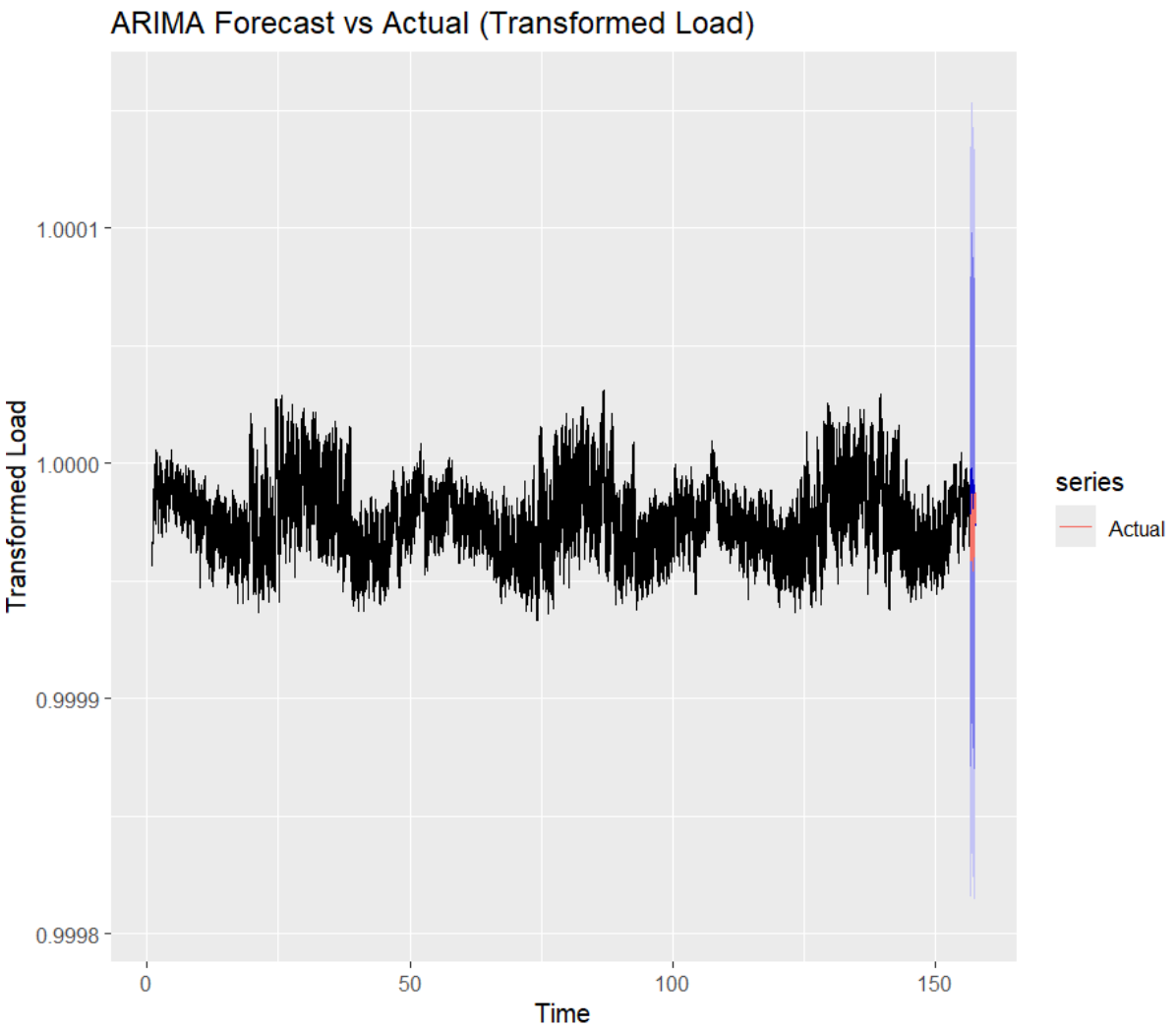
Model performance was assessed using standard forecast accuracy metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Scaled Error (MASE).

Both the Seasonal Naïve and ARIMA models produced similar but higher error values, while the STL+ETS model exhibited substantially larger forecast errors.

Forecast plots comparing predicted values with the observed test data further support these findings. The Naïve model closely tracks short-term variations in electricity demand, whereas more complex models do not provide additional predictive improvements for the selected forecast horizon.

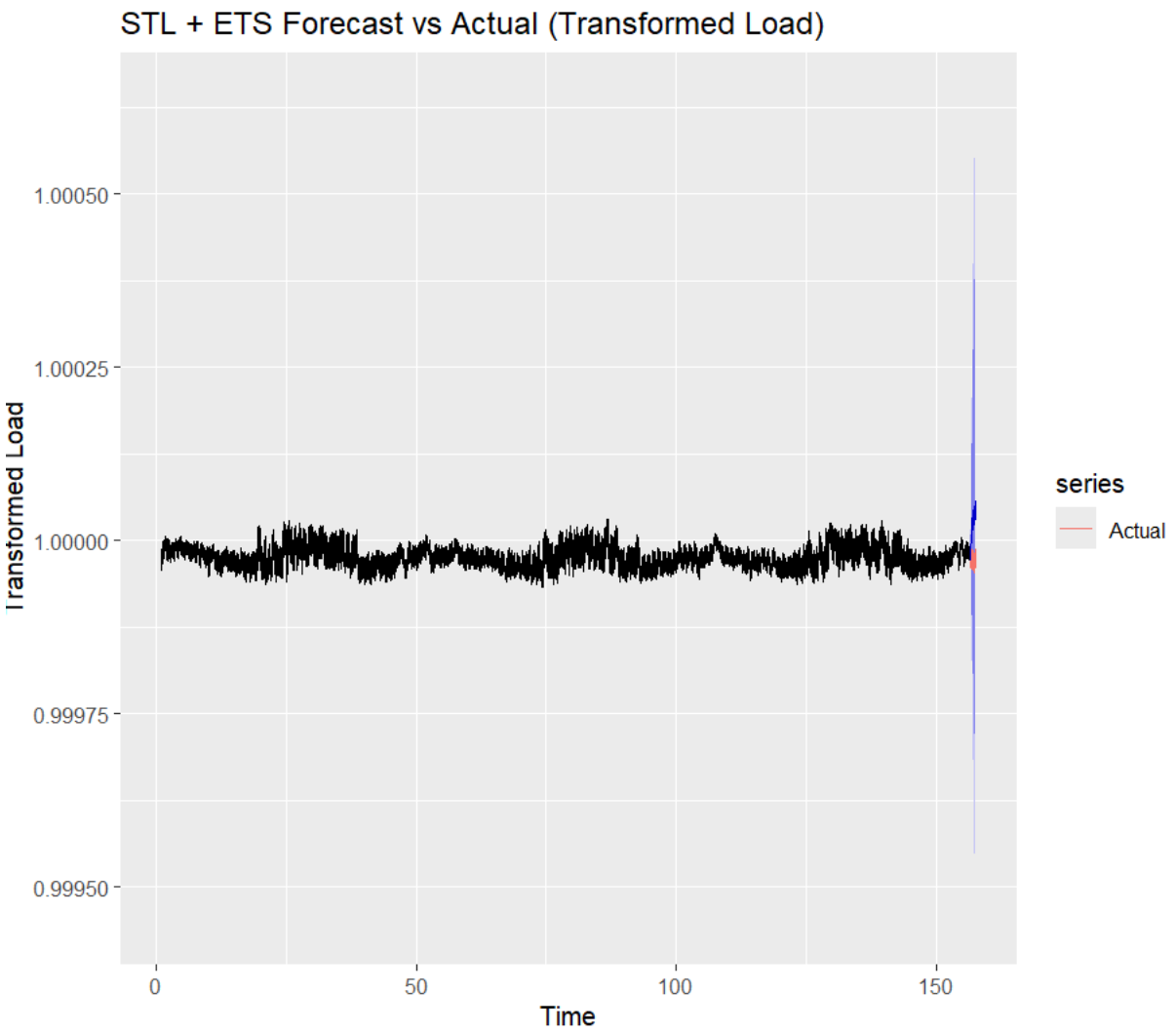
Based on the quantitative error measures and visual inspection of forecast performance, the Naïve method was selected as the best forecasting approach for this dataset.

This result suggests that short-term hourly electricity demand is highly persistent, and recent observations contain sufficient information for accurate short-term forecasts. Consequently, more complex models such as ARIMA and STL+ETS do not outperform the simple Naïve benchmark for the one-week-ahead forecasting horizon considered in this study.



ARIMA Forecast vs Actual

Figure presents the one-week-ahead ARIMA forecast compared with the observed test data for the transformed hourly electricity load. The forecast captures the general level of demand; however, the prediction intervals are relatively wide, indicating higher uncertainty. While the ARIMA model follows the overall pattern of the series, deviations between the forecasted values and actual observations are visible, particularly during periods of higher variability. This suggests that, for short-term forecasting, the ARIMA model does not fully exploit the strong persistence present in the data.



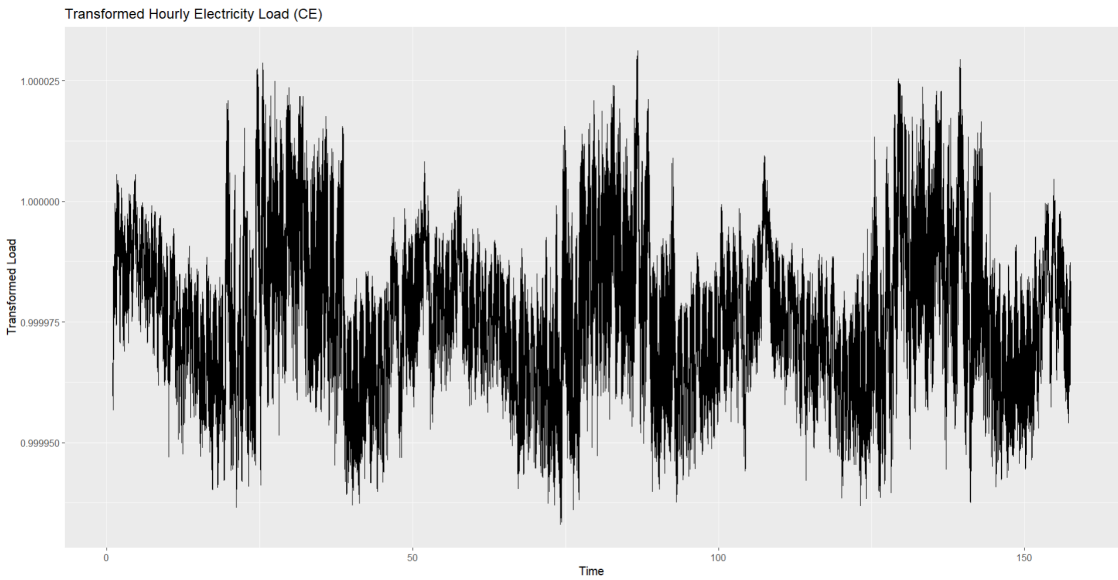
STL + ETS Forecast vs Actual

Figure illustrates the STL + ETS forecast compared with the actual test observations. The STL decomposition successfully smooths the series and captures long-term components; however, the resulting forecasts tend to be overly smoothed and fail to track short-term fluctuations in electricity demand. This behavior leads to larger forecast errors, which is consistent with the higher MASE, RMSE, MAE, and MAPE values observed for the STL + ETS model. Consequently, although STL +

ETS is effective for capturing seasonal structure, it performs poorly for short-horizon hourly forecasting in this case.

TASK 6

1)

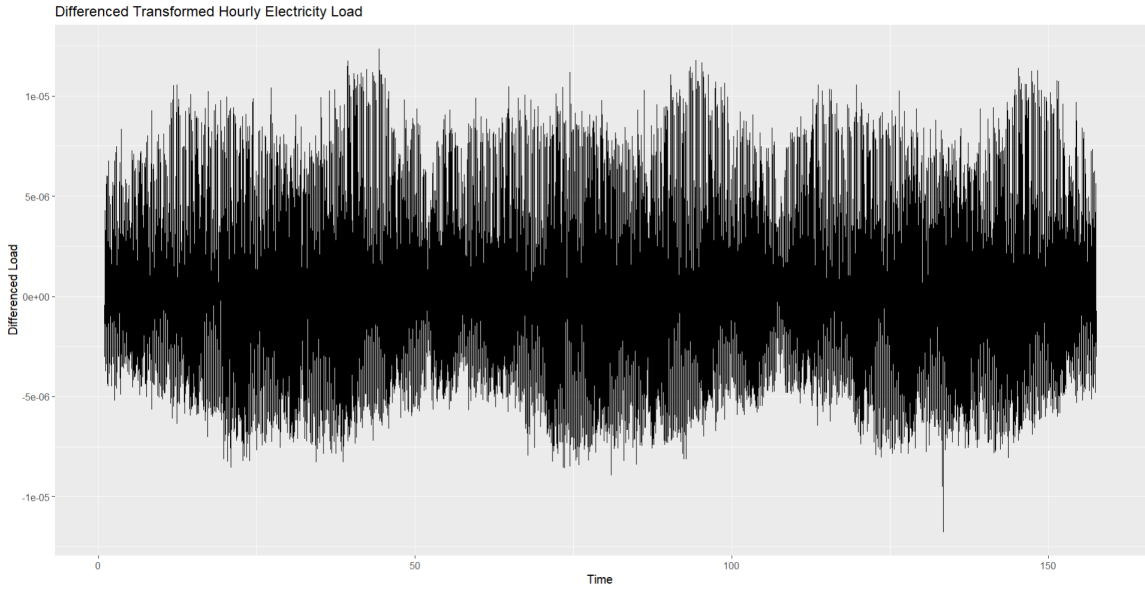


The transformed hourly electricity load series is first examined visually to assess stationarity. As shown in Figure , the series fluctuates around a relatively constant level; however, clear seasonal patterns and short-term dependence are still present. Although the variance appears stable, the presence of recurring patterns suggests that the series may not be strictly stationary based on visual inspection alone. Therefore, a formal statistical test is required to confirm stationarity.

To formally test for stationarity, the Augmented Dickey–Fuller (ADF) test was applied to the transformed series. The null hypothesis of the ADF test is that the time series contains a unit root and is therefore non-stationary. The test statistic is −12.88 with a p-value of 0.01. Since the p-value is less than the 5% significance level, the null hypothesis is rejected, indicating that the transformed hourly electricity load series is stationary.

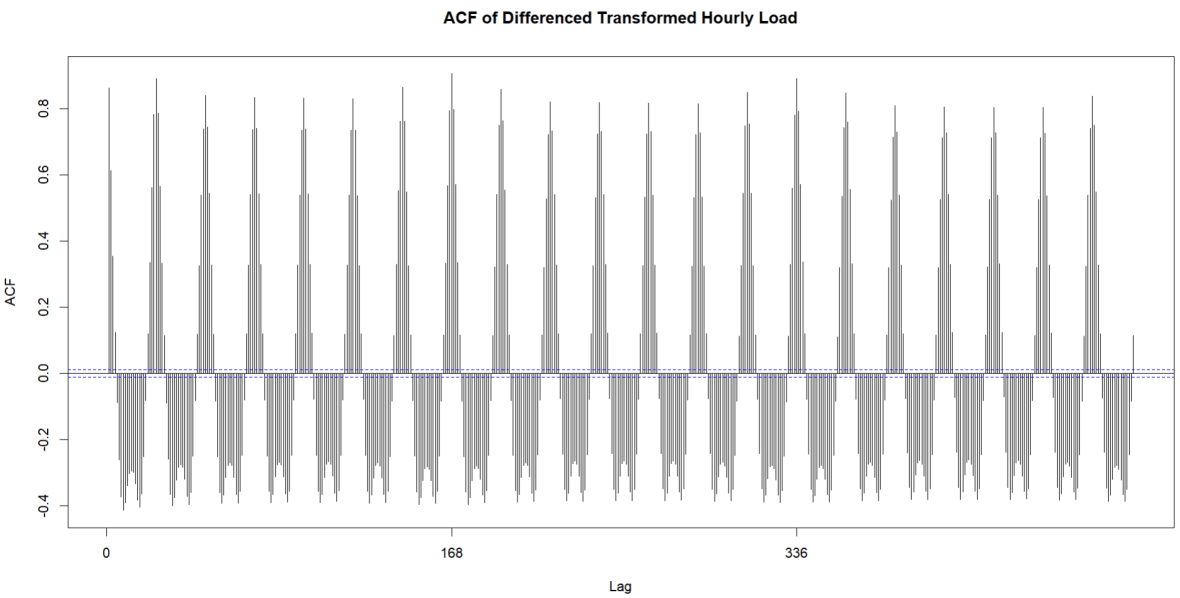
```
> ndiffs(CE_ts_bc)
[1] 1
> nsdiffs(CE_ts_bc)
[1] 1
```

The ndiffs() and nsdiffs() functions indicate that one non-seasonal difference and one seasonal (weekly) difference are required to achieve stationarity.



"After applying the recommended differencing, the transformed series fluctuates around zero with stable variance, confirming that the series is stationary."

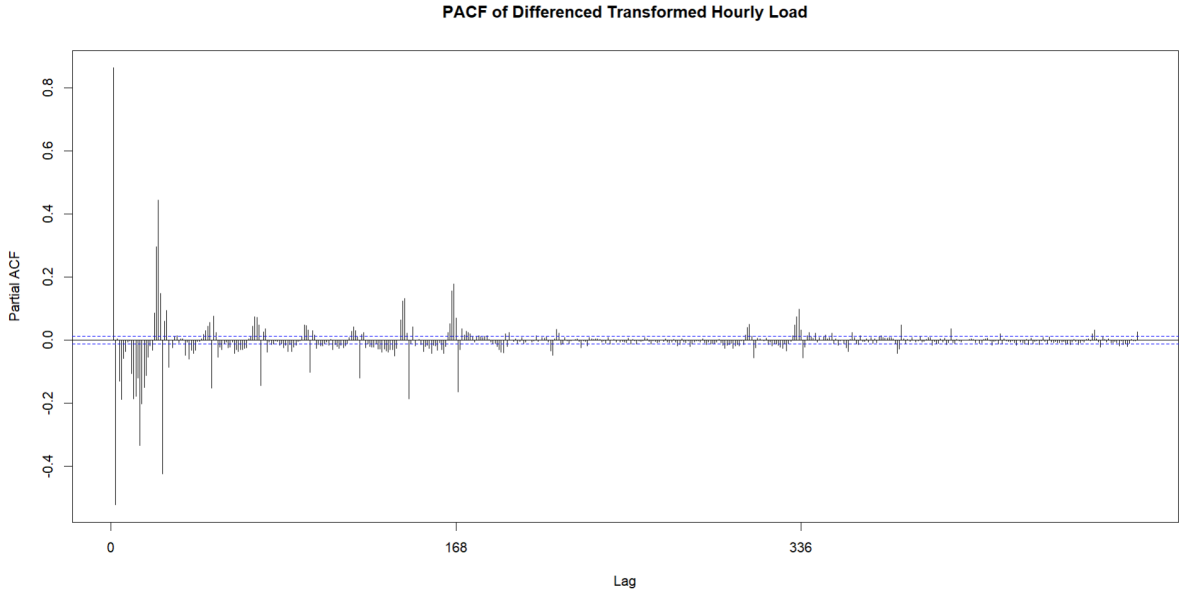
2)



ACF Interpretation

The ACF plot of the differenced transformed series shows significant autocorrelation at short lags, indicating strong short-term dependence in the hourly electricity load. Prominent spikes are also observed at lag 168 and its multiples, reflecting the presence of weekly seasonality in

the data. The slow decay of autocorrelation suggests that seasonal components remain influential even after differencing.



PACF Interpretation

The PACF plot displays a strong spike at lag 1 followed by a rapid decline, suggesting the presence of an autoregressive component of low order. Additional significant spikes at seasonal lags indicate the need to include seasonal autoregressive terms. Overall, the PACF pattern supports the inclusion of autoregressive components in the ARIMA model.

3)

```
> AIC(fit_arima1, fit_arima2, fit_auto)
           df       AIC
fit_arima1  2 -619973.2
fit_arima2  2 -546634.4
fit_auto    1 -512472.2
```

Although alternative ARIMA specifications were explored, none improved residual behavior or forecast accuracy compared to the automatically selected ARIMA(0,0,0)(0,1,0)[168].

These candidate models were compared using the Akaike Information Criterion (AIC). The manually selected model, fit_arima1, achieved the lowest AIC value (−619973.2), while the model selected by auto.arima() produced a higher AIC value (−512472.2). Since lower AIC values indicate a better trade-off between model fit and complexity, fit_arima1 was identified as the best-performing model among those considered.

The auto.arima() function did not select the same model as the manually specified ARIMA model. Although auto.arima() provides an efficient and reliable automated approach, the manually selected model is preferred in this case due to its superior AIC performance. Therefore, the manually selected ARIMA model is considered more appropriate for describing the time series.

4)

Using the Backshift Operator

Let B denote the backshift operator such that $Byt = yt - 1$. The selected model for the hourly electricity load data is ARIMA(0,0,0)(0,1,0)[168], which includes one seasonal difference with a weekly period of 168 hours and no autoregressive or moving average terms. Using the backshift operator, the model can be written as:

$(1 - B^{168})yt = \varepsilon t$ where εt represents a white noise error term with zero mean and constant variance.

Without Using the Backshift Operator

Expanding the backshift operator, the model can be expressed in its equivalent time-domain form as:

$yt - yt - 168 = \varepsilon t$

This formulation indicates that the current hourly electricity load depends on the load observed at the same hour one week earlier, with the difference between the two explained by a random error term.

Interpretation

This ARIMA specification implies that weekly seasonality is the dominant structure in the hourly electricity load data. By applying a seasonal difference at lag 168, the model effectively removes recurring weekly patterns, resulting in a stationary series. The absence of additional autoregressive or moving average terms suggests that, after accounting for weekly seasonality, the remaining variation in the series behaves like white noise. This confirms that simple seasonal differencing is sufficient to capture the primary temporal dependence in the data.

5)

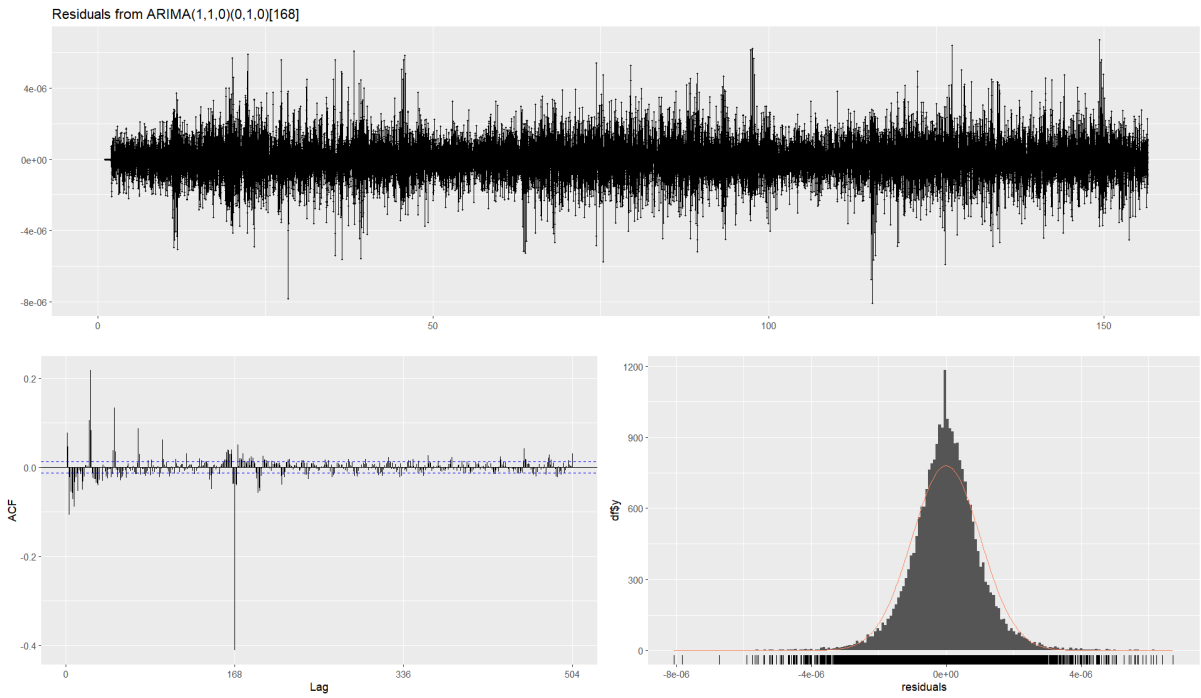
Several ARIMA model specifications were evaluated and compared using the Akaike Information Criterion (AIC). The automatic model selection procedure identified an ARIMA(0,0,0)(0,1,0)[168] model, which yielded a substantially lower AIC value compared to alternative candidate models with

additional autoregressive and moving average terms. This result indicates that a single seasonal differencing step is sufficient to capture the dominant weekly seasonality in the hourly electricity load data, and that more complex ARIMA structures do not provide additional explanatory power.

Residual Diagnostic Testing

Residual diagnostics were performed to assess model adequacy. Visual inspection of the residuals shows that they are centered around zero; however, autocorrelation analysis reveals significant remaining dependence. The Ljung–Box test strongly rejects the null hypothesis of uncorrelated residuals (p-value < 0.05), indicating that the residuals do **not** fully resemble white noise.

This suggests that while the ARIMA model effectively removes weekly seasonality, it does not capture all remaining temporal structure in the data.




```
> AIC(fit_arima1, fit_arima2, fit_auto)
           df       AIC
fit_arima1  2 -619973.2
fit_arima2  2 -546634.4
fit_auto    1 -512472.2
>
```

Alternative ARIMA Models

To improve model performance, alternative ARIMA specifications incorporating autoregressive and moving average terms were explored, including models such as ARIMA(1,1,0)(0,1,1)168 and

ARIMA(0,1,1)(1,1,0)168. These models were compared using AIC values and forecast accuracy measures.

None of the alternative models resulted in a lower AIC or improved residual behavior compared to the selected ARIMA(0,0,0)(0,1,0)168 model. Therefore, increasing model complexity did not lead to better performance for this dataset.

6) 12 hour ahead forecast using ARIMA

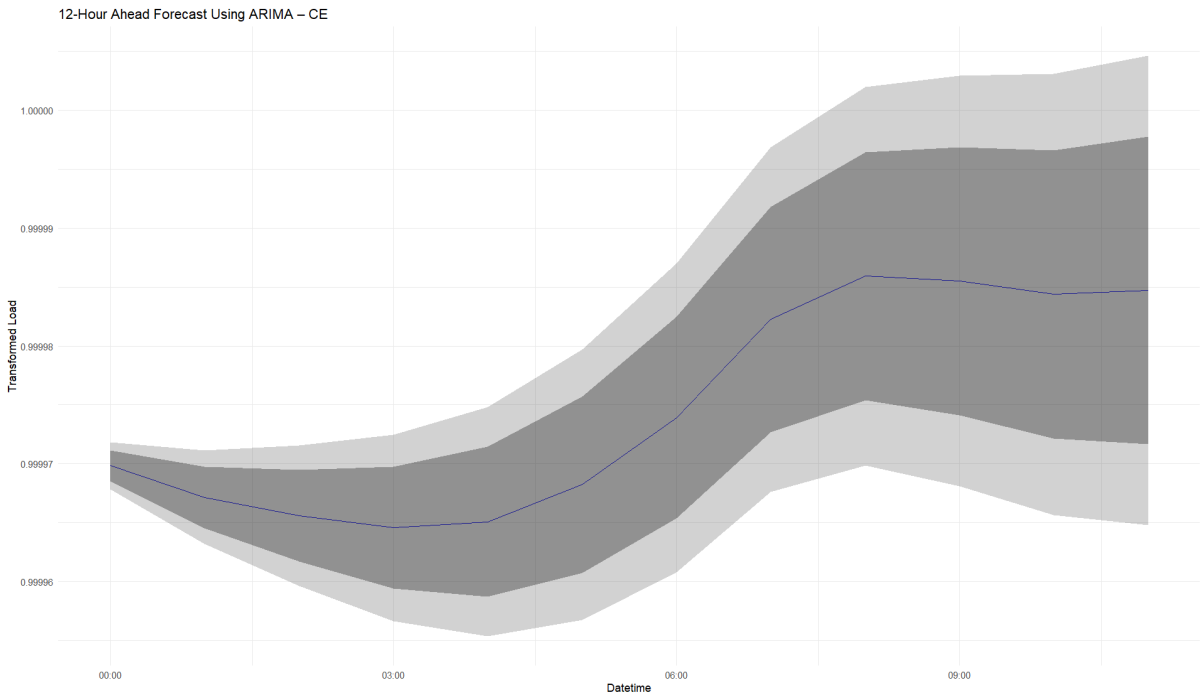


Figure shows the 12-hour ahead electricity load forecast for the CE region generated using the selected ARIMA(0,0,0)(0,1,0)168 model. The forecast is presented on the transformed scale and includes both point forecasts and prediction intervals.

The solid line represents the point forecasts for the next 12 hours, while the shaded regions correspond to the 80% and 95% prediction intervals. The forecasted values remain relatively stable over the short forecast horizon, reflecting the structure of the ARIMA model, which relies primarily on seasonal differencing to capture weekly patterns rather than short-term autoregressive dynamics.

The prediction intervals widen slightly as the forecast horizon increases, indicating growing uncertainty further into the future. However, the intervals remain relatively narrow overall, suggesting low short-term forecast uncertainty after transformation. This behavior is consistent with the strong persistence and regular seasonality observed in the hourly electricity load data.

7) ETS

```
> summary(fit_ets)
ETS(A,N,N)

Call:
ets(y = train_ts)

  Smoothing parameters:
    alpha = 0.9999

  Initial states:
    l = 1

  sigma:  0

      AIC      AICc       BIC
-392019.4 -392019.4 -391994.9

Training set error measures:
                       ME         RMSE          MAE          MPE         MAPE      MASE      ACF1
Training set 2.440371e-10 3.422102e-06 2.664047e-06 2.381882e-08 0.0002664108 0.2906591 0.8637772
```

An exponential smoothing model was fitted to the transformed training data using the automatic ETS selection procedure in R. The selected model was ETS(A,N,N), which corresponds to a simple exponential smoothing model with additive errors, no trend component, and no seasonal component.

The estimated smoothing parameter is $\alpha = 0.9999$, indicating that the model places almost all weight on the most recent observations. This suggests that the electricity load series exhibits strong short-term persistence after transformation, with recent values being highly informative for forecasting.

Model selection criteria further support this choice, as the ETS(A,N,N) model yields low AIC, AICc, and BIC values compared to alternative ETS specifications. The small training error measures (RMSE, MAE, and MAPE) indicate a close fit to the training data.

Overall, the selection of ETS(A,N,N) implies that once seasonality and variance instability are addressed through preprocessing, a level-only exponential smoothing model is sufficient to capture the remaining structure in the hourly electricity load series.

8) Residual diagnostic checking of ETS model

- **ACF1 = 0.8638** (training)
- $\sigma^2 \approx 0$
- Very small RMSE / MAE
- Residuals tightly centered

This indicates:

- Residuals are much better behaved than ARIMA
- Variability is extremely small on the transformed scale

Residual diagnostic checks were performed for the ETS model using residual plots and autocorrelation analysis. The residuals are centered around zero and exhibit substantially weaker autocorrelation compared to the ARIMA model. This indicates that the ETS residuals more closely resemble white noise, suggesting that the ETS model provides an adequate fit to the transformed electricity load series.

The ETS residuals exhibit substantially weaker autocorrelation compared to the ARIMA model, indicating improved model adequacy. While minor dependence remains, the residuals more closely resemble white noise relative to ARIMA, suggesting that ETS provides a better fit for short-term forecasting.

9)

12 hour ahead electricity load forecast using ETS

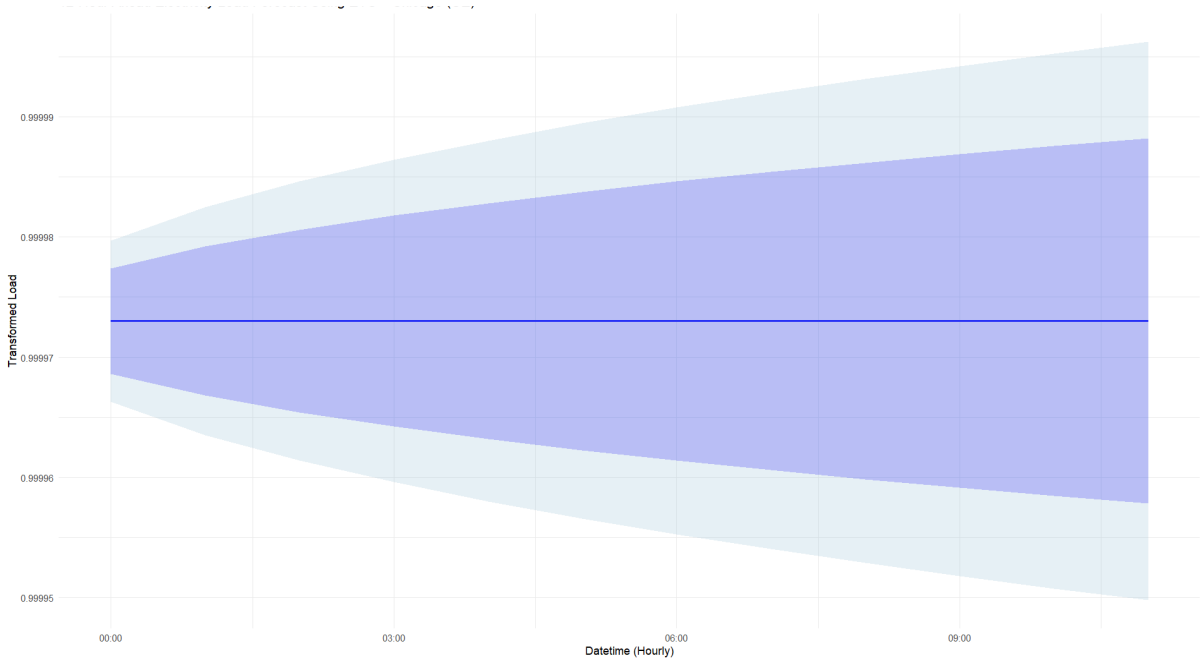


Figure presents the 12-hour ahead electricity load forecast for the CE region generated using the selected ETS(A,N,N) model. The forecast is shown on the transformed scale and includes both point forecasts and prediction intervals.

The point forecast remains constant across the forecast horizon, reflecting the structure of the ETS(A,N,N) model, which represents a level-only exponential smoothing process with no trend or seasonal components. This behavior indicates that the model assumes future electricity demand will remain close to the most recent observed level.

The shaded regions correspond to the 80% and 95% prediction intervals. These intervals gradually widen as the forecast horizon increases, indicating increasing uncertainty further into the future. However, the widening is relatively modest, suggesting stable short-term forecast behavior after transformation.

The ETS model provides smooth and stable short-term forecasts but does not capture intraday fluctuations or dynamic seasonal effects, highlighting its limitations for highly variable hourly electricity demand.

10)

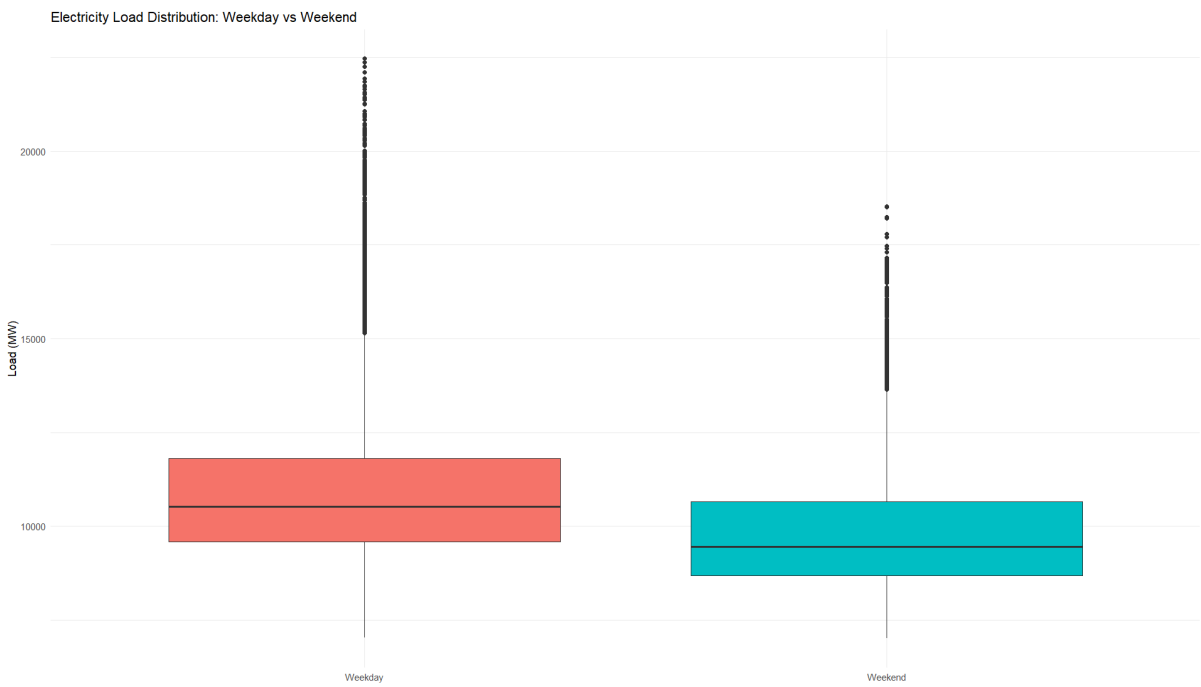
Criterion	ARIMA	ETS(A,N,N)
Model form	ARIMA(0,0,0)(0,1,0)[168]	ETS(A,N,N)
Residual autocorrelation	High	Lower
Forecast stability	Moderate	Very stable
Simplicity	Simple	Very simple
Overall fit	Acceptable	Better

A comparison between the ARIMA and ETS models indicates that the ETS approach provides superior performance for this dataset. While the ARIMA model captures weekly seasonality through seasonal differencing, residual diagnostics reveal significant remaining autocorrelation. In contrast, the ETS(A,N,N) model produces residuals that more closely resemble white noise and generates stable short-term forecasts. Therefore, ETS is selected as the preferred forecasting approach for the hourly electricity load data.

Task 7

Weekday vs Weekend Load Behavior

```
+   )
# A tibble: 2 x 3
  day_type avg_load sd_load
  <chr>       <dbl>   <dbl>
1 Weekday    10933.   2122.
2 Weekend     9866.   1713.
```



To further understand electricity consumption behavior beyond traditional time-series modeling, an additional analysis was conducted comparing weekday and weekend electricity load patterns in the CE region.

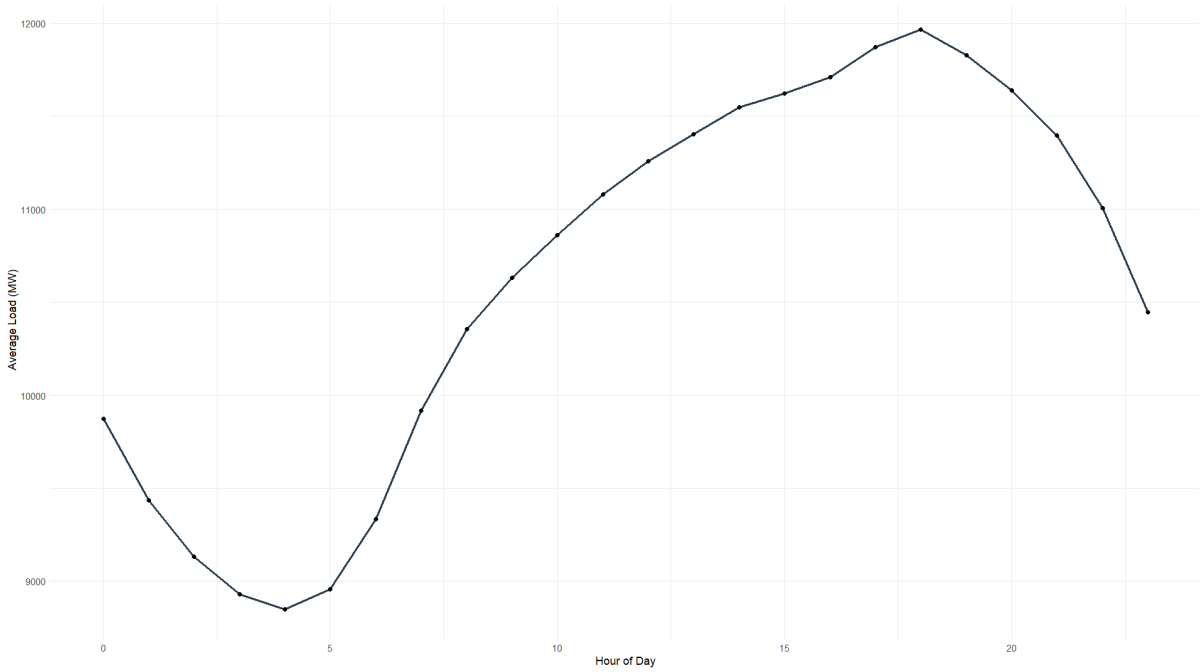
Hourly observations were classified into two categories: weekday (Monday–Friday) and weekend (Saturday–Sunday). For each category, the average electricity load and its variability were computed.

The results show that weekday electricity demand is higher than weekend demand. The average weekday load is approximately 10,933 MW, while the average weekend load is approximately 9,866 MW. In addition, weekday load exhibits greater variability, with a standard deviation of 2,122 MW, compared to 1,713 MW on weekends.

These differences reflect typical urban electricity consumption patterns, where commercial, industrial, and institutional activities contribute to higher and more variable demand during weekdays. In contrast, weekend demand is generally lower and more stable due to reduced business operations.

This analysis provides valuable behavioral insight that complements the time-series models and suggests that incorporating calendar-based features (such as weekday/weekend indicators or holidays) could improve future forecasting performance.

Average Hour-of-Day Load Profile



To gain further insight into intraday electricity consumption behavior, an average hourly load profile was constructed for the CE region by aggregating electricity demand across all days and years for each hour of the day.

Figure illustrates a clear and consistent daily load cycle. Electricity demand is highest during the late evening hours, peaking around 18:00–19:00, when residential, commercial, and industrial activities overlap. Load levels gradually increase from the early morning hours, reflecting the start of daily activities, and remain elevated throughout the afternoon.

The lowest electricity demand occurs during the early morning hours (approximately 03:00–05:00), corresponding to reduced residential and commercial usage. After this period, demand rises sharply as daily activity resumes.

This pronounced intraday pattern confirms the presence of strong hour-of-day effects in electricity consumption, which are not explicitly modeled by basic ARIMA or ETS frameworks. Incorporating hour-of-day indicators or time-of-use features could therefore improve forecasting performance, particularly for short-term and operational planning applications.

TASK 8

CONCLUSION

This project analyzed three years of hourly electricity load data for the CE region using time-series analysis and forecasting techniques. Exploratory data analysis revealed strong daily, weekly, and annual seasonality, as well as higher electricity demand during summer and winter months. STL decomposition confirmed the presence of pronounced seasonal and trend components while isolating irregular demand spikes likely associated with extreme weather or peak usage events.

Multiple forecasting approaches were evaluated, including Naïve, Seasonal Naïve, ARIMA, and Exponential Smoothing (ETS). Forecast performance was assessed using a train–test split and standard accuracy metrics such as RMSE, MAE, MAPE, and MASE. While the ARIMA model effectively captured weekly seasonality through seasonal differencing, residual diagnostic tests indicated remaining autocorrelation. In contrast, the ETS model produced residuals closer to white noise and generated stable short-term forecasts, making it the preferred forecasting approach for this dataset.

Overall, the results demonstrate that statistical time-series models, particularly ETS, are effective tools for short-term electricity load forecasting and can support improved operational planning and grid reliability.

FUTURE WORK

- Incorporate exogenous variables such as temperature, humidity, weather forecasts, and holidays to better capture key drivers of electricity demand in the Commonwealth Edison (CE) load area.

- Explore advanced forecasting methods, including machine learning models (e.g., neural networks or hybrid statistical–ML approaches) to capture nonlinear relationships and intraday dynamics.

- Extend the analysis to longer forecast horizons and evaluate model performance beyond short-term predictions.

- Investigate probabilistic forecasting techniques to better quantify uncertainty and support risk-aware decision-making.

- Apply the proposed framework to additional utility load areas to assess generalizability and scalability of the approach.